BTH545 Workshop 2

Learning Outcomes

- Create multiple widgets
- Use signals and slots to connect widgets together
- Use simple box layouts to arrange widgets

Development (60%)

For this lab, you need to create a Qt application which will display a slider configured to select values from 0 to 10 and a label beside it that shows the value from the slider and that is updated as the slider is moved. On the line below the slider, you will display a second label that displays the value from the slider as the word for the number selected by the slider. For example, if the slider is set to 6, then the label beside it will show "6" while the label underneath the slider will show "six". These labels will be updated every time the slider changes the value.

Once you have confirmed that the program works as intended, you should add two push buttons to the second row, beside the textual display of the number. One push button will be marked with "+" and the other with "-". The first will increment the number displayed and update all of the displays as well as the position of the slider to reflect the new number. The minus button will do the same but decrement the displayed number instead of incrementing it. If the buttons cause the number to go over 10, it should wrap around to become 0 and if the number goes less than 0, it should wrap around and become 10.

Submission

Once you have the program written, you should deploy it with the required libraries so that it can be run on any windows/mac system. Attach the deployment folder to the Blackboard submission page.

Reflection (40%)

Answer the following questions in a text file called reflect.txt and submit it along with the rest of your workshop. Your answers should be significant. The reflections should explore the topics in depth and not just give single line answers.

1. What is the function of the meta object compiler?
2. When you connect a signal to a slot, you use the syntax &QPushButton::clicked. Explain what this does.
3. When you create the push button explain how the parameters for the constructor determine that the button will appear on the desktop rather than as a child of another widget.
4. Explain what happens when you call the .show() method on the push button and give your opinion as to why the API was designed to require a show method.